

HALEEMA TALLAT

Machine Learning Engineer · AI evaluations, agents, LLM inference

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AI SAFETY & EVALUATIONS

LeakBench — github.com/haleemataallat/leakbench

2026

- Built a 30-task Inspect benchmark that tests whether AI agents can find and fix the bugs that make an ML evaluation report the wrong number, plus 5 correct pipelines that catch over-editing. Agents work in network-isolated Docker sandboxes and are scored by hidden checks.
- Made every task verifiable before any model ran: 120 no-model checks confirm each one is solvable and fails when the data is edited or the tests are faked. Found and fixed scoring ranges that rejected valid fixes or passed half-fixes.
- Showed it separates models: Qwen 3.5 9B solved 75% of bug tasks against 29% for the 4B (95% CI +29 to +61 points). Without a hint that the eval might be wrong, the 4B fell to 12%.

ConspiracyBench — Stationhouse, research collaborator · [Hugging Face](#)

2026

- Found the benchmark's headline divergence rate was inflated from about 20% to 49%: Qwen 3.5 4B used up its token budget on reasoning and returned empty answers on half its turns, and the LLM judge still graded them.
- Fixed it in the scorer (merged): empty answers are now marked invalid instead of judged, halving judge calls. Another contributor reproduced the issue on two more judge models.
- Built a log-analysis tool that shows where a run's time goes and checks byte for byte that speed changes leave prompts unchanged; it traced 51 timed-out requests per run to a concurrency default.

Coursework: Technical AI Safety, BlueDot Impact (2026) · ARENA curriculum, self-paced (2026)

EXPERIENCE

Senior AI Engineer, Agents & LLM Infrastructure — IgniteTech (via Crossover)

Jul 2025 – Present

- Designed and built IRIS's first production social engine in ~3 months, now handling 1,000+ customer interactions a day across 10–15 channels: multi-tenant Node.js and TypeScript services for Instagram, Facebook and LinkedIn ingestion, case processing and orchestration on AWS, plus NLP enrichment and React case-management features.
- Turned it into the template for the platform: its playbook and runbooks were reused for 20 more engines, and the platform now runs 27 on Kafka with KEDA autoscaling.
- Kept the pipelines up when downstream services failed, with circuit breakers, retries and Redis caching on Kafka consumers, and Prometheus metrics and Elasticsearch logs across services.
- Cut memory 6–10× and startup ~20× by replacing Java monoliths with stateless TypeScript services.
- Architected the team's agent platform, running up to 500 coding agents at once on LangGraph, Ray and Kubernetes, and made it measurable with Langfuse traces of every model and tool call and cost and error tracking in ClickHouse.
- Cut inference cost and latency with a vLLM and Anthropic API layer, and built the governed MCP layer for Claude Code, Cursor and internal agents, with human approval for high-impact actions.

Computer Vision Engineer (Contract) — Brick & Mortar AI

Mar – Oct 2025

- Showed a few-shot classifier reported at 46.7% actually scored 89.7% by tracing split leakage and protocol mismatches in its eval harness and rebuilding the held-out split.
- Cut cross-camera identity switches 20% and raised throughput 10–15× with segmentation-masked re-identification, served on Triton with TensorRT.

Computer Vision Engineer — NextBridge

Nov 2024 – Mar 2025

- Built a fail-safe WebRTC control link for a smart wheelchair, with a watchdog that stops the chair on link loss.

Machine Learning Engineer — Powersoft19

Nov 2023 – Nov 2024

- Cut field engineers' problem-resolution time 90% with a production hybrid RAG system over technical documents, measured with recall@k.
- Cut fault-detection turnaround from ~48 hours to ~30 minutes by leading 3 engineers to replace manual inspection with anomaly detection on 1TB+ of device telemetry.

Research Engineer — KICS, Center for Language Engineering, UET

Nov 2022 – Nov 2023

- Reached 93% accuracy on enterprise text classification and built an archive-scale OCR pipeline for TV news tickers.

EDUCATION & AWARDS

BSc Electrical Engineering, UET Lahore (2023) · CureMD Master Inventor Award (2022) · COHACK winner (2022)

SKILLS

Evaluations: Inspect, benchmark and task design, Docker sandboxes, LLM judges, paired bootstrap, Langfuse

LLMs & agents: LangGraph, Ray, MCP, vLLM, Hugging Face, Anthropic API, RAG, Qdrant, fine-tuning

Infrastructure: AWS, Kubernetes, Docker, Kafka, KEDA, Redis, PostgreSQL, ClickHouse, Elasticsearch, Airflow

Languages & ML: Python, TypeScript, Node.js, React, SQL, C++, PyTorch, CUDA, ONNX, TensorRT, Triton